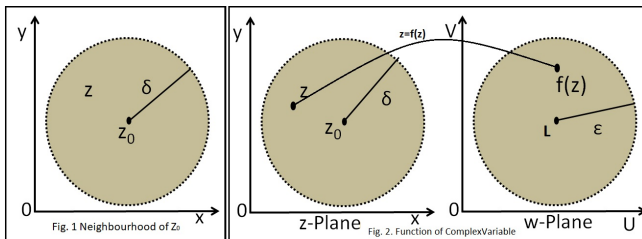


Neighbourhood of a point z_0

Let z_0 be a complex number and δ be a positive real number. Then the set of all points z satisfying $|z - z_0| < \delta$ is called a Neighbourhood(nbd). Thus, $N_\delta(z_0) = \{z : |z - z_0| < \delta\}$. we observe that, $|z - z_0| < \delta$ represent the interior of the circle with center Z_0 and radius δ .



Functions of Complex Variables

If for each value of the complex variable $z = x + iy$ in a given region $\mathbb{R}$, we have one or more values of $w = u + iv$, the w is called function of the complex variable z and we write $w = f(z) = u(x, y) + iv(x, y)$ where u and v are real functions of x and y .

Limit Point

A point z_0 is said to be a limit point of a set if every neighbourhood of z_0 contains infinitely many points.

Limit Point of a Function

Let $f(z)$ is a single valued function in a region $\mathbb{R}$ and let z_0 be a limit point of $\mathbb{R}$. Then, L is said to be the limit of $f(z)$ at z_0 if for every $\varepsilon > 0$, there exists a positive δ such that $|f(z) - L| < \varepsilon$ whenever $|z - z_0| < \delta$

$$\text{i.e., } \lim_{z \rightarrow z_0} f(z) = L.$$

Continuity of a Function

Let $f(z)$ is a single valued function in a region $\mathbb{R}$ and let z_0 be a limit point of $\mathbb{R}$. If

- ① $\lim_{z \rightarrow z_0} f(z)$ exists and
- ② $\lim_{z \rightarrow z_0} f(z) = f(z_0)$, then $f(z)$ is said to be continuous at $z = z_0$.

Example: Show that $f(x, y) = \frac{2xy}{x^2 + y^2}$ discontinuous at $(0, 0)$ given that $f(0) = 0$.

Soln: Given $f(x, y) = \frac{2xy}{x^2 + y^2}$.

$$\lim_{x \rightarrow 0}, \lim_{y \rightarrow 0} f(x, y) = \lim_{x \rightarrow 0} \left(\lim_{y \rightarrow 0} \frac{2xy}{x^2 + y^2} \right) = 0,$$

$$\lim_{x \rightarrow 0}, \lim_{y \rightarrow 0} f(x, y) = \lim_{y \rightarrow 0} \left(\lim_{x \rightarrow 0} \frac{2xy}{x^2 + y^2} \right) = 0.$$

Along the path $y = mx$, let $x \rightarrow 0$, $y \rightarrow 0$ simultaneously,

$$\begin{aligned} \lim_{y=mx}, \lim_{x \rightarrow 0} f(x, y) &= \lim_{y=mx}, \lim_{x \rightarrow 0} \frac{2xy}{x^2 + y^2} \\ &= \lim_{y \rightarrow 0}, \lim_{x \rightarrow 0} \frac{2mx^2}{x^2 + m^2x^2} = \lim_{y \rightarrow 0}, \lim_{x \rightarrow 0} \frac{2m}{1 + m^2} \neq 0. \end{aligned}$$

Hence, the function is not continuous at origin.

Example: Test continuity of $f(x, y) = \frac{2xy^2}{x^2 + y^4}$ at the origin given that $f(0, 0) = 0$.

Soln: Given $f(x, y) = \frac{2xy^2}{x^2 + y^4}$.

$$\lim_{x \rightarrow 0} \left(\lim_{y \rightarrow 0} \frac{2xy^2}{x^2 + y^4} \right) = 0, \quad \lim_{y \rightarrow 0} \left(\lim_{x \rightarrow 0} \frac{2xy^2}{x^2 + y^4} \right) = 0.$$

Along the curve $x = y^2$, let $x \rightarrow 0$, $y \rightarrow 0$ simultaneously,

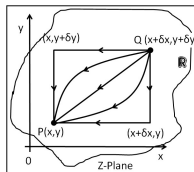
$$\lim_{x=y^2}, \lim_{x \rightarrow 0} \frac{2xy^2}{x^2 + y^4} = \lim_{y \rightarrow 0}, \lim_{x \rightarrow 0} \frac{2y^4}{2y^4} = 1 \neq 0.$$

Hence, the function is not continuous at origin.

Differentiability of a Point

Let $f(z)$ is a single valued function in a region $\mathbb{R}$ and let z_0 be a limit point of $\mathbb{R}$. Then, the function $f(z)$ is differentiable at z_0 , if the function continuous at z_0 and

$$f'(z) = \frac{dw}{dz} = \lim_{z \rightarrow z_0} \frac{f(z) - f(z_0)}{z - z_0} \text{ exists.}$$



Here, $P(z)$ and $Q(z + \delta z)$ be two neighbouring points . The point Q may approach P along straight line or curved path in the given region.

Note: for each value of z there corresponding one and only one value of w is called **single value function** (z^2 , $1/z$) of z , otherwise it is called a **multi valued function** ($\sqrt{z}$, $\sqrt[3]{z}$) of z .

Analytic or Holomorphic or Regular Function

A single valued function $f(z)$ is defined in a region $\mathbb{R}$ is said to be analytic at the point z_0 if $f(z)$ is differentiable at every point of some neighbourhood of z_0 .

Entire or Integral Function

A single valued function $f(z)$ is defined in a region $\mathbb{R}$ is said to be entire at the point z_0 if $f(z)$ is analytic at every point of some neighbourhood of z_0 . An entire function is analytic everywhere except at $z = \infty$.

Example: z^2 , e^z , $\cos(z)$, $\sin(z)$, $\cosh(z)$, $\sinh(z)$.

Note: A point at which the function $w = f(z)$ fails to be analytic is called a singular point or singularity of $f(z)$.

Any functions : Continuous $\rightarrow$ Differentiable $\rightarrow$ Analytic $\rightarrow$ Entire

Necessary condition for $f(z)$ to be analytic (Cauchy-Riemann Equations)

If $f(z) = u(x, y) + iv(x, y)$ is an analytic function in a region $\mathbb{R}$, then

$$\frac{\partial u}{\partial x} = \frac{\partial v}{\partial y} \text{ and } \frac{\partial v}{\partial x} = -\frac{\partial u}{\partial y}, \text{ i.e., } u_x = v_y \text{ and } v_x = -u_y.$$

Sufficient condition for $f(z)$ to be analytic

A single valued function $f(z) = u(x, y) + iv(x, y)$ defined in a region $\mathbb{R}$, if

- 1 The first order partial derivatives with respect to x and y exists in $\mathbb{R}$

$$\text{i.e., } \frac{\partial u}{\partial x}, \frac{\partial u}{\partial y}, \frac{\partial v}{\partial x} \text{ and } \frac{\partial v}{\partial y} \text{ exists and}$$

- 2 all these partial derivatives are continuous and
- 3 $u_x = v_y$ and $v_x = -u_y$ at every point in $\mathbb{R}$, then $f(z)$ is an analytic function in that region $\mathbb{R}$.

Polar Form of C-R Equations

Let (r, θ) denote the polar co-ordinates of the point (x, y) . Then

$$\frac{\partial u}{\partial r} = \frac{1}{r} \frac{\partial v}{\partial \theta} \text{ and } \frac{\partial v}{\partial r} = -\frac{1}{r} \frac{\partial u}{\partial \theta}, \text{ i.e., } u_r = \frac{1}{r} v_\theta \text{ and } v_r = -\frac{1}{r} u_\theta.$$

Harmonic Functions

A function $f(z)$ is said to be a harmonic function if it satisfy the Laplace equations

$$\frac{\partial^2 f}{\partial x^2} + \frac{\partial^2 f}{\partial y^2} = 0, \text{ i.e., } \nabla^2 f = 0.$$

Note: Let $f(z) = u(x, y) + iv(x, y)$ be an analytic function, then

- $u(x, y)$ is conjugate harmonic function of $v(x, y)$ and
- $v(x, y)$ is conjugate harmonic function of $u(x, y)$.

Example: Prove that the function $f(z) = z^2$ is analytic.

Soln: Let $f(z) = u + iv$, where $z = x + iy$

$$\Rightarrow u + iv = z^2 = (x + iy)^2 = x^2 + i^2 y^2 + i2xy = (x^2 - y^2) + i2xy.$$

Equating real and imaginary parts, we get

$$u = x^2 - y^2 \text{ and } v = 2xy$$

Diff. w. r. t. x and y , we obtain

$$u_x = 2x \text{ and } v_x = 2y,$$

$$u_y = -2y \text{ and } v_y = 2x.$$

From above equations,

$$u_x = v_y = 2x \text{ and } u_y = -v_x = -2y.$$

Hence, Cauchy-Riemann equations satisfied.

Therefore, $f(z) = z^2$ is analytic.

Example: Show that the following

(i) $f(z) = e^z$ (ii) $f(z) = \sin(z)$ are analytic.

Hint:

$$(i) e^{i\theta} = \cos\theta + i\sin\theta,$$

$$(ii) \sin(A+B) = \sin(A)\cos(B) + \cos(A)\sin(B),$$

$$\sin(i\theta) = i\sinh(\theta),$$

$$\cos(i\theta) = \cosh(\theta).$$

Example: Show that the function $f(z) = \log(z)$ is analytic everywhere except at the origin. Also, find its derivative.

Soln: Let $f(z) = u + iv$, where $z = re^{i\theta}$

$$\Rightarrow u + iv = \log z = \log(re^{i\theta}) = \log(r) + \log(e^{i\theta}) = \log(r) + i\theta. \quad \longrightarrow (1)$$

At the origin $r = 0$,

$$\Rightarrow f(z) = \log(0) + i\theta = -\infty + i\theta.$$

So, $f(z)$ is not analytic at origin. To show other than origin, equating real and imaginary parts, we have

$$u(r, \theta) = \log(r), \quad v(r, \theta) = \theta.$$

Diff. w. r. t. r and θ , we obtain

$$u_r = 1/r \text{ and } v_r = 0,$$

$$u_\theta = 0 \text{ and } v_\theta = 1.$$

From above equations,

$$u_r = (1/r)v_\theta = 1/r \text{ and } v_r = (-1/r)u_\theta = 0.$$

Hence, Cauchy-Riemann equations satisfied.

So, u_r , u_θ , v_r and v_θ are continuous everywhere except at origin. The function $f(z)$ satisfies all sufficient condition for existence of derivatives except at origin.

Therefore, $f(z) = z^2$ is analytic everywhere except at origin.

The derivatives is

$$f'(z) = \frac{u_r + iv_r}{e^{i\theta}} = \frac{u_r + i(0)}{e^{i\theta}} = \frac{1/r}{e^{i\theta}} = \frac{1}{re^{i\theta}} = \frac{1}{z}.$$

Example: Show that $f(z) = |xy|^{1/2}$ is not analytic at origin even though C-R equations are satisfied at the point.

Soln: Let $f(z) = u + iv$, where $z = x + iy$
 $\Rightarrow u + iv = |xy|^{1/2}. \quad \rightarrow (1)$

Equating real and imaginary parts, we get

$$u = |xy|^{1/2} \text{ and } v = 0.$$

Diff. w. r. t. x and y and substitute $x = 0, y = 0$ (at origin), we get

$$\frac{\partial u}{\partial x}_{(0,0)} = \lim_{x \rightarrow 0} \left[\frac{u(x, 0) - u(0, 0)}{x} \right] = \lim_{x \rightarrow 0} \left[\frac{(0 - 0)}{x} \right] = 0,$$

$$\frac{\partial u}{\partial y}_{(0,0)} = \lim_{y \rightarrow 0} \left[\frac{u(0, y) - u(0, 0)}{y} \right] = 0, \quad \frac{\partial v}{\partial x}_{(0,0)} = \lim_{x \rightarrow 0} \left[\frac{v(x, 0) - u(0, 0)}{x} \right] = 0 \text{ and}$$

$$\frac{\partial v}{\partial y}_{(0,0)} = \lim_{y \rightarrow 0} \left[\frac{v(0, y) - u(0, 0)}{y} \right] = 0.$$

Clearly, C-R equations are satisfied. That is, $u_x = v_y$ and $u_y = -v_x$ at origin.

$$\begin{aligned} \text{Now, } \lim_{z \rightarrow 0} \left[\frac{f(z) - f(0)}{z} \right] &= \lim_{x \rightarrow 0, y \rightarrow 0} \left[\frac{\sqrt{|xy|} - 0}{xy} \right] \\ &= \lim_{y=mx, x \rightarrow 0} \left[\frac{\sqrt{m|x|^2} - 0}{x(1+im)} \right] = \frac{\sqrt{m}}{(1+im)}. \quad [\because \text{Along } y = mx] \end{aligned}$$

The limit is not unique, since it depends on the value m . Therefore, $f'(z)$ does not exist.

Hence, $f(z)$ is not analytic at origin.

Example: Prove that $f(z) = \frac{x^3(1+i) - y^3(1-i)}{x^2 + y^2}$, ($z \neq 0$), $f(0) = 0$ is continuous and that C-R equations are satisfied at origin, yet $f'(z)$ does not exist there.

Soln: Let $f(z) = u + iv = \frac{x^3(1+i) - y^3(1-i)}{x^2 + y^2}$, where $z = x + iy$

$$\Rightarrow u + iv = \frac{(x^3 - y^3) + i(x^3 + y^3)}{x^2 + y^2}. \quad \longrightarrow (1)$$

Equating real and imaginary parts, we get

$$u = \frac{(x^3 - y^3)}{(x^2 + y^2)} \text{ and } v = \frac{(x^3 + y^3)}{(x^2 + y^2)}.$$

Diff. w. r. t. x and y and substitute $x = 0$, $y = 0$ (at origin), we get

$$\begin{aligned} \frac{\partial u}{\partial x}(0,0) &= \lim_{x \rightarrow 0} \left[\frac{u(x,0) - u(0,0)}{x} \right] = \lim_{x \rightarrow 0} \left[\frac{(x-0)}{x} \right] = 1, \\ \frac{\partial u}{\partial y}(0,0) &= \lim_{y \rightarrow 0} \left[\frac{u(0,y) - u(0,0)}{y} \right] = \lim_{y \rightarrow 0} \left[\frac{(-y-0)}{y} \right] = -1, \\ \frac{\partial v}{\partial x}(0,0) &= \lim_{x \rightarrow 0} \left[\frac{v(x,0) - u(0,0)}{x} \right] = \lim_{x \rightarrow 0} \left[\frac{(x-0)}{x} \right] = 1 \text{ and} \\ \frac{\partial v}{\partial y}(0,0) &= \lim_{y \rightarrow 0} \left[\frac{v(0,y) - u(0,0)}{y} \right] = \lim_{y \rightarrow 0} \left[\frac{(y-0)}{y} \right] = 1. \end{aligned}$$

Clearly, C-R equations are satisfied.

That is, $u_x = v_y = 1$ and $u_y = -v_x = -1$ at origin.

$$\text{Now, } \lim_{z \rightarrow 0} \left[\frac{f(z) - f(0)}{z} \right] = \lim_{x \rightarrow 0, y \rightarrow 0} \left[\frac{(x^3 - y^3) + i(x^3 + y^3)}{(x^2 + y^2)(x + iy)} \right]$$

Cont.

Along $y = 0$,

$$\Rightarrow \lim_{x \rightarrow 0} \frac{(x^3 + ix^3)}{(x^2 \cdot x)} = (1 + i).$$

Along $y = x$,

$$\Rightarrow \lim_{x \rightarrow 0} \frac{2ix^3}{2x^3(1 + i)} = \frac{i}{(1 + i)}.$$

Here, $f'(0)$ have different values for different curves. So, the limit is not unique and not exist.

Hence, $f'(0)$ is not a regular(analytic) function at the origin.

Examples:

- ❶ Verify whether $\frac{x - iy}{x^2 + y^2}$ is an analytic function.
- ❷ Determine whether the function with $u = x^2 - y^2$ and $v = \frac{-y}{x^2 + y^2}$ is an analytic function or not.
- ❸ Prove that $z^3 + z$ is an analytic function.
- ❹ Show that $f(z) = |z|^2$ is differentiable at origin but it not analytic.

Example: Prove that $u = e^x \cos(y)$ is harmonic.

Soln: Given $u = e^x \cos(y)$. $\rightarrow (1)$

We know that $\nabla^2 u = 0$, if u is harmonic.

Diff. (1) partially w.r.t, we have

$$\frac{\partial u}{\partial x} = e^x \cos(y) \Rightarrow \frac{\partial^2 u}{\partial x^2} = e^x \cos(y) \rightarrow (2)$$

$$\frac{\partial u}{\partial y} = e^x (-\sin(y)) \Rightarrow \frac{\partial^2 u}{\partial y^2} = -e^x \cos(y) \rightarrow (3).$$

(2) + (3) $\Rightarrow$

$$\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = e^x (\cos(y) - \cos(y)) = 0.$$

Hence, the given function u is harmonic.

Examples: Show that, the following functions are harmonic

❶ $u(x, y) = \frac{1}{2} \log(x^2 + y^2),$

❷ $u(x, y) = e^x (x \cos(y) - y \sin(y)),$

❸ $v(x, y) = 3x^2 y - y^3,$

❹ $v(x, y) = e^{-x} (2xy \cos(y) + (y^2 - x^2) \sin(y)).$